Research Proposal Form

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Centre name	Btec FPT Ha Noi
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Unit	Computing Research Project
Proposed title	

Section One: Title, objective, responsibilities

1**. Project Title**

Design and Implementation of a Zero Trust Web-Based Access Control System for Enterprise Security

2. Project Objectives

General Objective

To design and develop a web-based application based on the Zero Trust security model in order to secure enterprise systems by strictly controlling authentication, authorization, and access to internal resources.

3. Project Responsibilities

- Project Leader: Manage project progress and coordination
- Backend Developer: Implement authentication, authorization, and API security

Section Two: Reasons for choosing this research project

- Rising cybersecurity threats in enterprises
- Traditional security models are no longer effective
- Zero Trust provides a modern and practical security approach
- Strong relevance to real-world enterprise systems
- Rich academic and technical resources available

Section Three: Literature sources searched

Conducting a literature review is essential to establish theoretical and empirical foundations for research on Zero Trust Architecture (ZTA). Search Strategy: Academic databases such as SpringerLink, ScienceDirect, MDPI, and Google Scholar were used to collect peer-reviewed studies related to Zero Trust security and access control models. Key Search Terms: “Zero Trust Architecture”, “access control”, “identity-based authentication”, “least privilege access”, “continuous verification”, and “network micro-segmentation”.

Reputable Journals: Springer Cybersecurity, Computers & Security (ScienceDirect), Sensors (MDPI), and Electronics (MDPI). Representative References: • Springer (2024) – Research trends in Zero Trust Architecture • ScienceDirect (2023) – Zero Trust implementation factors • MDPI Sensors (2025) – Zero Trust in IoT environments • MDPI Electronics (2024) – Access control in Zero Trust systems These sources provide a strong academic foundation for developing a secure Zero Trust-based web platform.

Section Four: Research approach and methodologies

- **Literature Review:** Study academic papers, industry standards, and security frameworks related to the Zero Trust model.
- **Requirement Analysis:** Identify enterprise security problems and define system requirements based on Zero Trust principles.
- **System Design:** Design the system architecture, access control model, and security mechanisms.
- **Implementation:** Develop the web application using JWT authentication, RBAC authorization, and secure APIs.
- **Testing and Evaluation:** Test authentication, authorization, and security controls to evaluate system effectiveness.

Comments and agreement from tutor

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Comments and agreement from project proposal checker (if applicable)